

Machine learning estimation of battery state of health in residential photovoltaic systems

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ABSTRACT

As the global adoption of residential battery storage systems paired with local photovoltaic (PV) generation increases, prosumers are increasingly motivated to reduce both their electricity costs and dependence on the grid. This shift highlights the importance of accurately evaluating and predicting the battery's State of Health (SOH) and Remaining Useful Life (RUL). These factors are crucial for determining the operational costs and longevity of battery systems. Traditionally, SOH predictions have relied heavily on detailed measurement data and time-intensive simulations. In response, we introduce a new AI-based approach that simplifies SOH estimation. Our method, named "ML Battery Life Predictor (MLBatLife)," leverages forecasted or historical PV generation data and load consumption patterns to quickly forecast the SOH for various battery configurations. Tested on simulated data, this tool demonstrated a high accuracy, with a coefficient of determination of 0.986 for predictions one day ahead, and an impressively low average error of 0.1 % for projections five years into the future. This innovative AI-driven technique offers substantial benefits for evaluating the economic viability and warranty parameters of battery installations in different regions. It provides a valuable resource for both industry stakeholders and energy system planners aiming to assess and anticipate battery health outcomes efficiently.

Introduction

Advancements in battery storage systems (BSS) lead to a rapid evolution in residential photovoltaic (PV) systems and therefore, transforming energy systems (a list of all acronyms is provided in Appendix E). These advances not only improve the reliability of these systems but also make them to be economically feasible. It is obvious that a key driver for this transformation is a reduction in the costs of solar panels and BSSs [1], which results in an exponential increase in the installation of these systems. As reported by Bloomberg New Energy Finance (BNEF), the worldwide installed capacity of these systems exceeded 30 GW by the end of 2021. Similarly, according to [2], Germany has been embracing growth in these systems installations, accounting for over 1.1 million units installed and a combined capacity surpassing 10 GWh by January 2024. As expected, a combination of residential PV and BSSs becomes more essential for the future sustainable development in

the energy sector.

Battery aging affects both performance and financial returns of BSSs over time, which translates in the economic feasibility of residential PV systems. A novel method for PV-based battery energy storage systems proposed in [3] highlights the critical role of battery aging and technology choice in maximizing economic viability through cost-optimal sizing and operation patterns. The pivotal factors in determining battery lifetime are site-specific conditions as well as storage dispatch strategies [4]; thereby influencing the economic outcomes of residential BSSs. Moreover, Munzke et al. [5] reveals that factors such as battery aging, inverter efficiency, and charging strategies designed to mitigate calendar aging crucially affect the cost-effectiveness of PV home storage systems. In detail, being less sensitive to calendar aging can reduce the costs around 20 % and underscore the financial advantages of minimizing battery degradation. Likewise, Vonsien and Madlener [6] indicate that the key factor for the cost-effectiveness BSSs is its lifetime,

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which is mainly dominated by state-of-charge limits. A real design and techno-economic assessment in [7] provides a detailed examination of operational strategies to mitigate the effects of battery aging; it presents a nuanced analysis that improves understanding of battery degradation mechanisms and the economic feasibility of energy storage solutions. These findings emphasize the critical relevance of accounting for battery aging in the economic evaluation of BSSs and propose solutions for maximizing lifespan and economic performance.

The prediction of SOH, particularly lithium-ion batteries, has received a lot of attention since it plays such an important role in guaranteeing the dependability, safety, and efficiency of battery systems in a variety of applications. Methods for SOH prediction can be roughly classified as model-based, data-driven, or hybrid (a combination of model-based and data-driven techniques), with each having advantageous and disadvantageous, which are described below:

Model-based methods (white box models)

These approaches are based on electrochemical and physical principles and aim to mathematically describe the battery's behavior and degradation mechanisms. They often require fundamental knowledge of the battery's internal chemical processes, which can be complex and computationally intensive. The review by Spotnitz and Franklin [8] discusses the basic mechanisms of lithium-ion battery degradation. These methods are praised for their explanatory power but criticized for their high demand for computational resources and detailed model parameters, which may not be readily available. Alternatively, battery behavior can be modelled by a simpler equivalent electric model, as it is described in [9,10].

Data-driven methods (black box models)

These methods predict SOH based on historical data using machine learning without explicitly modeling the battery's internal chemical processes. These methods can adapt to various battery types and conditions and offer flexibility and scalability. Recent advancements have seen the application of deep learning techniques, such as Deep Neural Networks (DNNs) [11] and Recurrent Neural Networks (RNNs) [12,13], for SOH prediction; which demonstrate promising results in terms of accuracy and robustness. However, the "black box" nature of such models and the substantial data requirements for training, poses challenges in interpretability and data availability, respectively.

Hybrid Methods (gray box models)

These approaches utilize both model-based insights and data-driven flexibility to enhance prediction accuracy and reliability. These methods aim to mitigate the limitations associated with purely model-based or data-driven approaches. Additionally, they offer a balanced solution for SOH prediction. For instance, fusion technology methods are emerging as a powerful tool for improving SOH estimation and prediction accuracy [14].

Challenges in SOH prediction span across all categories. They include the need for extensive and high-quality data for data-driven methods, the complexity and computational demands of model-based approaches, and the integration complexity of hybrid methods. Furthermore, ensuring the generalizability of SOH prediction models across different battery types, operating conditions, and aging mechanisms remains a significant hurdle. One key factor influencing the performance of batteries is temperature, as highlighted in numerous studies [15–21].

In recent years, notable developments can be seen in battery health management thanks to machine learning techniques. As can be seen in Table 1, generally, the research trend in this area can be divided into cell- and system-levels. Cell-level studies highlight how important it is to have cell-specific prediction to monitor the battery health. For example, Tian et al. [22] and Hemdani et al. [23] employ advanced deep-learning

Table 1

ML methods being used to predict battery SOH.

Cell/ system level	Paper	Use case (operation of battery)	Input data	Comments
Cell	Tian et al. [22]	Random Walk (RW) Operation	NASA	Uses online sequential learning for efficient SOH prediction
	Hemdani et al. [23]	EVs Prognostics and Health Management	Experimental Measurement	Deep learning for low-error SOH prediction
	Liang et al. [27]	Real vehicle applications	NASA	LSTM-SVR model leveraging indirect health features from operational data by using a nonlinear constraint optimization
	Hou et al. [28]	BMS	CALCE	Applies Two-stage TrAdaBoost.R2 algorithm with transfer learning
	Akbar et al. [29]	EVs	Experimental Measurement	Web service for SOH testing in real-world EV scenarios
	Zhang et al. [30]	EVs	NASA	PSO-ELM model for accurate SOH and RUL estimation
	Liang et al. [31]	EVs	Experimental Measurement	RF model with PSO optimization for enhanced SOH prediction
	Chen et al. [32]	EVs	From [33]	Neural network approach for capacity estimation across battery life
	Zhang et al. [34]	EVs	NASA	SSA-DELM method for high-accuracy SOH prediction
	Jia et al. [35]	Safe operation of BESS	NASA	Sparrow search optimized deep learning for robust SOH prediction
System	Yuan et al. [36]	Battery health management	NASA	Two-Layer Stacking regressor model for accurate SOH and RUL prediction
	Benlamine et al. [37]	Battery prognostics of EVs	Massachusetts Institute of Technology (MIT)	Analysis of degradation factors and prediction approach on MIT datasets
	Shu et al. [24]	EVs safe operation	CALCE	LSTM network with transfer learning for data-efficient SOH prediction
	Battaglia et al. [25]	BMS	Impedance measurements (EIS)	Random Forest (RF) method is used for

(continued on next page)

Table 1 (continued)

Cell/ system level	Paper	Use case (operation of battery)	Input data	Comments
	Jafari et al. [26]	EVs	Taken from [38]	accurate and reliable SOC and SOH prediction XGBoost algorithm with accuracy correction for improved SOH estimation
	Yang et al. [39]	EVs	Real-World Vehicle Data (i. e. can be available upon request)	LSTM network focusing on resistance data for SOH prediction
Both	Hu et al. [40]	EVs Battery Management System (BMS) and Battery Second Life Applications	NASA, CALCE, and A123 (MIT and Stanford)	Gaussian process regression for accurate SOH estimation

models to predict SOH with acceptable precision according to cell-level input datasets. Having a precise SOH prediction is crucial to e.g. approve the demanding requirement specifications on longevity for automotive or stationary battery use cases. System-level studies instead provide a more comprehensive picture of battery health within different operational conditions in an application context and may be useful to e.g. check warranty statements on remaining useful life for a full battery storage system. Some of system-level studies are listed in Table 1. For example, Shu et al. [24] and Battaglia et al. [25] use the Long Short-Term Memory (LSTM) network and Random Forest method, respectively, to integrate broad system features into their SOH prediction. Moreover, Jafari et al. [26] employ XGBoost as an advanced data analysis method to improve the accuracy of the system's SOH prediction. These studies provide insight into the importance of having system-level datasets for accurate and reliable SOH predictions.

Following in the footsteps of previous research, we explore how to incorporate system-level data into residential PV systems and identify unique capabilities and associated challenges. Unlike previous studies that typically focused on either system-wide interactions or cell-level specification, our work incorporates system-level factors to predict SOH with practical realism on residential conditions. Our goal is to provide a comprehensive and accurate understanding of battery health with respect to specific application profiles for self-consumption increase applications using PV generation and residential electricity demand profiles. This approach not only places our work in the forefront of SOH prediction research, but also creates a unique niche by combining detailed system-level insights with the practical requirements of residential energy systems, thereby providing an elegant framework for SOH prediction.

The main contribution of this work is to show that ML techniques can provide a very convenient way (accurate and computationally efficient) to predict the expected life of battery systems operating in real conditions. The possibility to have an almost instantaneous estimation of the battery life is a key factor for the economic analysis in many real-world applications.

The paper is organized as follows: in Section 2, the proposed methodology and the employed datasets are described; the main results of increasing-complexity prediction models are explained in Section 3; Section 4 summarizes and compares the outcomes of the different models, analyzes its computational costs and proposes a physical explanation of these results; and finally, the main conclusions of the research are presented in Section 5.

Additionally, the more specific aspects of the research are detailed in appendixes: describing the features used to characterize power profiles (Appendix A); showing the results of preliminary simple models (Appendix B); and explaining abnormal results of high-dimensional models (Appendix C). As a further outcome of this research, an ML Battery Life Predictor has been made openly available [41].

Methodology

Dataset description

To assess the capability of Machine Learning (ML) techniques in predicting SOH, we considered a real-world scenario involving the application of BSSs to augment Self Consumption Increase (SCI) of electricity in residential settings. Specifically, we analyzed the power demand of 74 families, as recorded over the course of one year (2010) by HTW Berlin [42]. Concurrently, the power output from a typical rooftop photovoltaic (PV) installation was measured under actual operating conditions for the same duration in the preceding year (2009) [43]. Both datasets were captured at a high resolution of one sample per second.

From these data, we computed the residual power for each household, which represents the difference between PV generation and residential demand. The dynamics of these three signals—PV generation, user demand, and residual power—are illustrated in Fig. 1 for a typical household on a sunny day. The figure shows the power generated by the PV system (green line) on a sunny day, the power demand of a typical household (red line), and the residual power (blue line, representing the difference between the two) fluctuating throughout the day. When the residual power is positive (indicated by the green shaded area), the battery is charging; conversely, when it is negative, the battery is discharging.

To simulate long-term battery operations, each one-year residual power profile collected was replicated five times to create a five-year signal. This extended signal served as the input for a battery simulator using the SimSES simulation framework [44], an open-source Python-based tool developed by the Institute for Electrical Energy Storage Technology at the Technical University of Munich, which utilizes an equivalent electric circuit model for its simulations (gray box model). The concept of using battery digital twins for training ML degradation models is also explored, as reported in [45].

The aging model used in SimSES considers that the capacity degradation in battery storage systems is a result of two distinct processes: calendar aging and cycle aging. As detailed in [46], calendar aging can be modeled using the following expression:

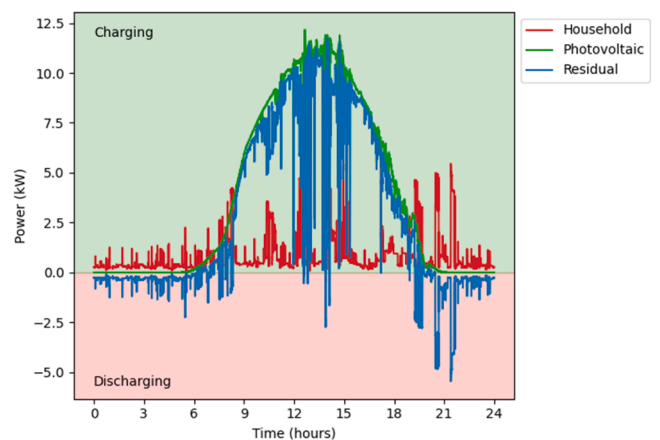


Fig. 1. Evolution of PV generation (green), household demand (red), and residual power (blue) for a typical household during a sunny day. The sign of residual power indicates when the battery is being charged (in the green region) or discharged (in the red region).

$$Q_{\text{loss}}^{\text{cal}}(t) = K_T^{\text{cal}} K_{\text{SOC}}^{\text{cal}} \sqrt{t}, \quad (1)$$

While calendar aging is influenced by factors like ambient temperature and SOC, represented by the parameters K_T^{cal} and $K_{\text{SOC}}^{\text{cal}}$ respectively, cycle aging follows a different mechanism. According to [47], cycle aging can be modeled by the following expression:

$$Q_{\text{loss}}^{\text{cyc}}(t) = K_T^{\text{cyc}} K_{\text{Crate}}^{\text{cyc}} K_{\text{DOD}}^{\text{cyc}} \sqrt{\text{FEC}}, \quad (2)$$

Cycle aging is affected by the number of full equivalent charging-discharging cycles (FEC). The parameters influencing this process include K_T^{cyc} , which depends on the ambient temperature, $K_{\text{Crate}}^{\text{cyc}}$, which is influenced by the C-rate (the charge or discharge current with respect to the battery's nominal capacity), and $K_{\text{DOD}}^{\text{cyc}}$, which depends on the depth of discharge (DOD) of the battery. Consequently, the total capacity degradation of the battery can be expressed as:

$$Q_{\text{loss}}(t) = K_T^{\text{cal}} K_{\text{SOC}}^{\text{cal}} \sqrt{t} + K_T^{\text{cyc}} K_{\text{Crate}}^{\text{cyc}} K_{\text{DOD}}^{\text{cyc}} \sqrt{\text{FEC}}. \quad (3)$$

On the other hand, the number of full equivalent charging-discharging cycles (FEC) increases over time and can be represented by the function $\text{FEC} = f_{\text{FEC}}(t)$. While this is generally a non-linear function, it can be approximated by the first term of its Taylor expansion as $\text{FEC} \approx K_{\text{FEC}} t$. Consequently,

$$Q_{\text{loss}}(t) \approx \left(K_T^{\text{cal}} K_{\text{SOC}}^{\text{cal}} + K_T^{\text{cyc}} K_{\text{Crate}}^{\text{cyc}} K_{\text{DOD}}^{\text{cyc}} \sqrt{K_{\text{FEC}}} \right) \sqrt{t} = K_Q \sqrt{t}. \quad (4)$$

For the simulation, each of the 74 input profiles (residual power signals) was processed under two distinct operation strategies: a greedy strategy and a feed-in damping strategy. In the greedy strategy, the residual power is directly routed to the BSS until it is either fully charged or discharged. This approach can lead to undesired power demand peaks from the grid, or excess power fed into it [48]. To mitigate such peaks, a damping strategy on the feed-in is employed, which incorporates forecasts of solar production and user demand to modulate the power exchange with the grid [49,50].

The battery cell used in all simulations was a Lithium-Ion Battery (LIB) with a Lithium-Iron-Phosphate (LFP) cathode and a Carbon-Graphite (C) anode, a nominal capacity of 2850 mAh and a nominal voltage of 3.2 V.

The output from the battery simulation is an SOH profile that details the five-year evolution of the battery system, maintaining the same one-second resolution as the input profiles. This output specifically represents the normalized remaining capacity (denoted as Q) of the BSS. To describe it more formally, for the i -th household the output profile is defined as follows:

$$\text{SOH}^{(i)}(t) = 1 - Q_{\text{loss}}^{(i)}(t) = 1 - \frac{Q(0) - Q^{(i)}(t)}{Q(0)} = \frac{Q^{(i)}(t)}{Q(0)}, \quad (5)$$

where $Q^{(i)}(t)$ represents the capacity at time t , $Q(0)$ is the initial capacity of the BSS at the beginning of its life, and $Q_{\text{loss}}^{(i)}(t)$ denotes the capacity that has been lost up to time t , relative to the initial capacity $Q(0)$.

By tracking the SOH at the beginning and the end of each day, we derive an incremental SOH profile that captures daily changes in capacity and resistance over a five-year period. This approach allows for a detailed analysis of the battery's performance and degradation over time. Formally, the incremental SOH for the i -th profile on the k -th day is defined as follows:

$$\Delta \text{SOH}^{(i)}(k) = \text{SOH}^{(i)}(t_k) - \text{SOH}^{(i)}(t_{k-1}), \quad (6)$$

where t_k is the time at the end of the k -th day.

To ensure sufficient data for training the ML models, input profiles are segmented into one-day lengths—a choice that will be justified in Section 2.2. Consequently, each input or output profile comprises 1825 segments, calculated as 365 days per year multiplied by 5 years. Overall, the complete experimental dataset includes 135,050 segments, which

results from multiplying the 1825 segments per household by the 74 households involved in the study.

As an alternative to direct battery simulation, the change in SOH (i.e., $\Delta \text{SOH}^{(i)}(k)$) for each profile can be estimated using machine learning (ML) techniques (black box model). Given that the input profile segment for a single day comprises 86,400 values (24 hours $\times$ 3600 seconds per hour), using this raw data directly in an ML model would require a dataset size beyond the available instances [51]. To address this, the characterization of each segment is condensed into a set of features that statistically summarize the time-domain or frequency-domain aspects of the data. For this study, we have adopted a generic set of 29 features as proposed in [52], which are detailed in Appendix A.

Consequently, the dataset employed in the ML approach is organized into a design matrix X with $n = 270,100$ rows (corresponding to 135,050 segments across two operational strategies) and $d = 29$ columns (one for each feature). The target variable, ΔSOH , is represented by a column vector with 270,100 entries.

The estimated values of $\widehat{\Delta \text{SOH}}^{(i)}$ over several consecutive days can be cumulatively added to provide an estimation of the SOH by the end of the k -th day.

$$\widehat{\text{SOH}}^{(i)}(k) = \text{SOH}^{(i)}(0) + \sum_{u=1}^k \widehat{\Delta \text{SOH}}^{(i)}(u). \quad (7)$$

Finally, the value of SOH at any continuous time t , occurring during the k -th day, can be estimated by interpolating between the SOH values at the beginning and the end of that day. This interpolation provides a smooth estimation of SOH across any point in time during the day, allowing for a detailed and dynamic assessment of battery health.

$$\widehat{\text{SOH}}^{(i)}(t) = \widehat{\text{SOH}}^{(i)}(k-1) + \frac{\widehat{\text{SOH}}^{(i)}(k) - \widehat{\text{SOH}}^{(i)}(k-1)}{t_k - t_{k-1}}(t - t_{k-1}), \quad (8)$$

where $t_k - 1 \leq t \leq t_k$.

The performance of the ML approach is evaluated by comparing the estimated values of $\text{SOH}(t)$ and $\Delta \text{SOH}(k)$ against those obtained from the simulation, which serves as the ground-truth. This comparison helps to validate the accuracy and reliability of the ML models in replicating the battery's state of health as predicted by the simulation. The various concepts involved in this research, including data segmentation, feature extraction, ML estimation, and performance comparison, are visually summarized in the diagram shown in Fig. 2.

Segment length

An essential consideration in constructing the dataset is the decision regarding the length of the segments. Shorter segments could significantly increase the number of instances available for training the ML model, potentially enhancing its performance. However, excessively short segments might fail to capture the typical behavior patterns of residual power adequately. Given the daily periodicity of solar energy and the routine daily activities in a household, selecting a one-day length for each segment appears to be a reasonable and intuitive choice. This length is likely the shortest span that can still reliably represent the daily dynamics of energy consumption and production.

This initial intuition regarding the one-day segment length is substantiated by a spectral analysis of the one-year input profile (subsequent years are repetitions of the first), as illustrated in Fig. 3 for a typical household. The spectrum reveals a principal harmonic with a period of one day and a secondary significant component at a twice-per-day frequency (12-hour period). This pattern is consistent across various household datasets, as confirmed by similar findings reported in [53]. The absence of significant harmonics for periods longer than one day indicates that all relevant information is encapsulated within one-day length segments. This empirical evidence underpins our choice of segment length as detailed in Section 2.1.

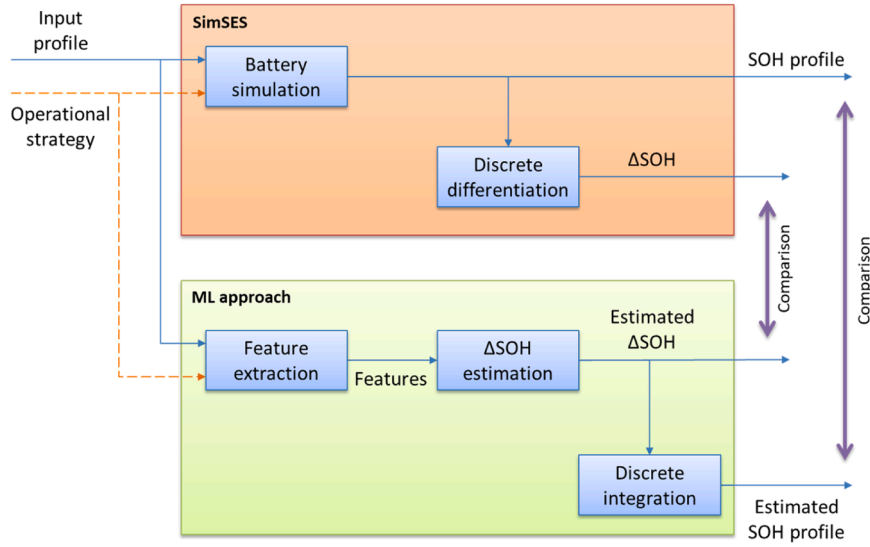


Fig. 2. Processing of input profiles: based on an electric-equivalent model simulation (top); and using a Machine Learning approach (bottom).

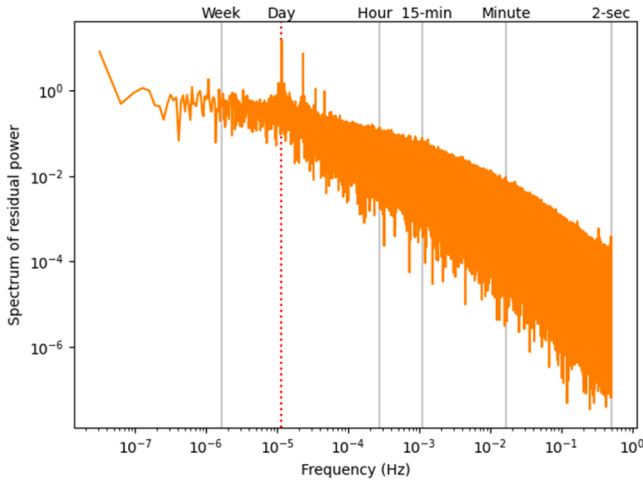


Fig. 3. Spectrum of a one-year residual power for an example of input profile. Red dashed line indicates the segmentation of 1 day as is conducted for subsequent ML analysis.

Machine learning regressor

Given that the values of ΔSOH are continuous, the machine learning approach to estimate these values necessitates the use of a regression model rather than a classifier. Among the various algorithms available, the Random Forest regressor has been chosen for several compelling reasons. Firstly, Random Forests often yield superior results, even outperforming deep neural networks, particularly when the available training dataset is not extensive enough [54]. Secondly, previous research using the same datasets has demonstrated that Random Forest achieved the most accurate results [55].

Random Forest regressors, which are an ensemble of regression trees, have been utilized to significantly reduce overfitting to the training data and minimize the generalization error—essentially the model's estimation error when analyzing new, unseen data [56]. In this study, we employed the `scikit-learn` implementation of Random Forests [57]. While the default settings for the model's hyperparameters were maintained in most cases, we adjusted two key parameters to optimize performance: the number of trees in the forest and the maximum depth of the trees.

To optimize the parameters and hyperparameters of the Random

Forest (RF), the 148 input profiles (comprising 74 residual power signals across two operational strategies) were randomly divided into three subsets: about 60 % (92) were allocated to the training dataset to tune the RF parameters; 20 % (28) were set aside for validation, which facilitated the selection of the best hyperparameters; and the remaining 20 % (28) constituted the testing set, used to estimate the generalization error of the model.

The primary metric used to evaluate the performance of the ML models is the coefficient of determination, commonly denoted as R^2 [58]. This statistical measure is used to quantify the proportion of variance in the dependent variable that is predictable from the independent variables. For a set of n actual values, y , and their corresponding estimated values, $\hat{y}$, the coefficient of determination is defined as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad (9)$$

Here, $\bar{y}$ is the mean of the actual values y . For a perfect regressor, R^2 would equal 1, indicating that the model explains all the variability of the response data around its mean. Conversely, for the simplest regressor—one that always predicts the mean value, $\bar{y}$, regardless of the input— R^2 would be 0, indicating that the model does not explain any of the variability of the response data around its mean.

The values of the RF hyperparameters were carefully selected by exploring various combinations through a grid search, an example of which is illustrated in Fig. 4. This exploration demonstrated that increasing the maximum depth beyond 100 and the number of trees beyond 50 does not significantly enhance model performance. Consequently, these values were chosen as the optimal hyperparameters, achieving an R^2 of 0.7234. Further increases in hyperparameter values, such as using 100 trees with a maximum depth of 10,000, resulted in a negligible improvement in regression performance (R^2 of 0.7260) while significantly increasing computational demands. This method of hyperparameter optimization was applied consistently across the different models proposed in the study.

Dimensionality reduction

The residual power input profiles are characterized using a set of features detailed in Appendix A. These features provide generic statistical summaries that capture both time-domain and frequency-domain

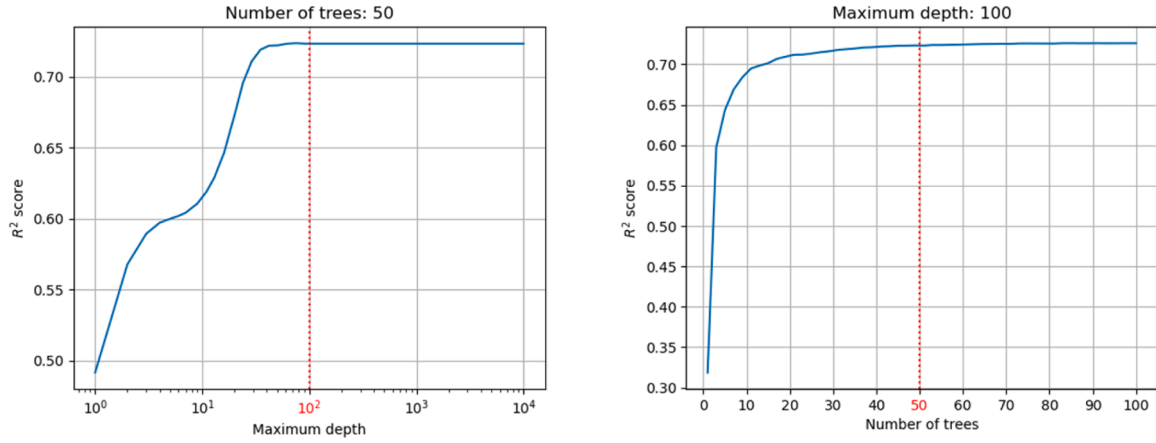


Fig. 4. Example of a random forest hyperparameter selection. A maximum depth of 100, and a set of 50 trees are selected as no significant improvement is obtained with greater values.

characteristics of the residual power. While some of these features are essential for predicting the State of Health (SOH), others might be redundant or irrelevant. On the one hand, the inclusion of non-contributive attributes can degrade the performance of the models [59]. On the other hand, employing a larger ensemble of features can reduce the explainability of the ML models and their predictions, making it harder to interpret the underlying factors driving the results [60].

To enhance both performance and explainability of our ML models, we are investigating the possibility of reducing the number of features utilized. Initially, each input profile is represented by a vector x_i in the original design matrix, with a dimensionality of $d = 29$. To simplify the model, these vectors can be transformed into a reduced dimensionality space. In this low-dimensional space each input profile is characterized by a vector z_i defined by d_z components, with $d_z < d$. The most straightforward method to achieve this goal is by feature selection, where the most important (d_z) features are identified and retained based on their significance to the estimation task. This involves evaluating

each feature's contribution to the outcome and discarding the less impactful ones.

Several feature selection algorithms are well-documented in the literature [61]. In this study, we have implemented two distinct approaches. Initially, we utilize the feature importance metric generated as a byproduct of the Random Forest regression algorithm. This method provides an importance metric for each feature at no additional computational cost [62]. While this method successfully ranks the features by importance, it does not specify how many features can be safely discarded without significantly affecting the model's predictive performance.

To address this limitation, we also employed a sequential forward feature selection algorithm [63]. This technique begins with an empty feature set and progressively adds features, one at a time, selecting the feature that maximizes model performance at each step. This iterative process continues until all possible features have been evaluated, resulting in an ordered set of features each associated with a specific

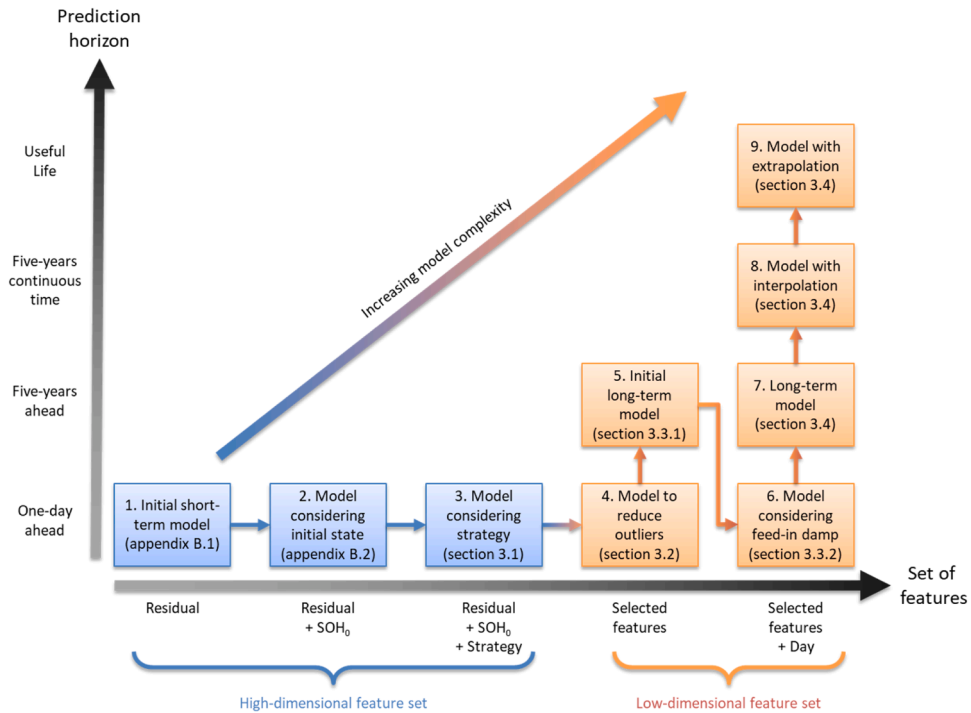


Fig. 5. Summary of the increasing-complexity prediction methods applied in this research: vertical axis indicating prediction horizon; horizontal axis specifying complexity of the process to obtain the set of features.

performance metric based on the subset's length.

Results

The methodologies outlined in the previous section have been applied to predict the SOH evolution of a BSS based on the input profiles contained in our dataset. Initial findings provided valuable insights for refining the predictive procedures and expanded the scope to include long-term predictions. Consequently, we explored up to nine models of increasing complexity, which are detailed throughout this section. Fig. 5 visually summarizes these models, depicting an increasing complexity of the feature extraction process on the horizontal axis, while the vertical axis illustrates progressively challenging prediction horizons. Models 1–3, highlighted in blue, employ a high-dimensional description of the segments. In contrast, models 4–9, shown in orange, utilize a reduced set of features for characterizing input profiles. For clarity, the section of the article discussing each model is indicated in Fig. 5. Models 1 and 2 are further explained in Appendix B for detailed reference.

In presenting the results, we initially focus on the overall capacity loss of the BSS. Subsequently, distinctions between predictions of calendar aging and cycle aging are discussed towards the end of this section.

High-dimensional short-term capacity estimation

The initial models explored in this study focus on high-dimensional characterization of residual power, as detailed in Section 2.1. These models are designed to predict short-term (one-day ahead) degradation of the SOH.

Model #1, represented in Fig. 5, utilizes a basic set of features that exclusively considers the residual power. The methodology and results of this model are thoroughly detailed in Appendix B.1. This model operates under the assumption that daily capacity loss remains constant regardless of the BSS's absolute capacity, suggesting a linear degradation over time. However, this assumption does not hold up under scrutiny.

Fig. 6 (left panel) displays the five-year evolution of BSS capacity for a randomly selected household under two different operational strategies, clearly showing non-linear behavior and variable daily capacity loss.

The capacity degradation of the BSS does not follow a linear pattern but instead exhibits a behavior that can be approximated by a square root function, as demonstrated in the right panel of Fig. 6 when the horizontal axis is scaled to the square root. This observation aligns with the physical dynamics of Solid Electrolyte Interface growth, a primary factor in calendar aging [64]. Moreover, the deviation from a perfect

straight line across different seasons indicates less degradation during the winter months, highlighting seasonal variations in the degradation process.

To account for the observed non-linearity in capacity degradation, a modification was made to the design matrix used in Model #1. Specifically, the capacity at the beginning of the day (SOH₀) was added as a new 30th column, forming the basis of Model #2, as illustrated in Fig. 5. The comprehensive results and analysis of this enhanced model are detailed in Appendix B.2.

While Models #1 and #2 effectively utilize the residual power profile for predictions, they do not differentiate between the operational strategies of the BSS—"greedy" and "feed-in damp"—which are pivotal for accurate modeling. To incorporate this aspect, the operational strategy was added to the design matrix (residual + SOH₀ + strategy) as a new 31st column, creating Model #3 in Fig. 5. Detailed outcomes from this further expanded model are provided in Appendix B.3.

Low-dimensional short-term capacity estimation

The final High-Dimensional Random Forest (HDRF) model used for short-term capacity prediction (Model #3) generally delivers excellent results. However, a few outliers have been identified, prompting a detailed examination detailed in Appendix D, reaching the conclusion that the random forest algorithm incorporates physically meaningless features. To address this issue, a preliminary feature selection process was implemented using a forward feature selection algorithm, as detailed in Section 2.4.

The results of this process are displayed in Fig. 7, where the incremental contribution of each new feature to the regression performance (R^2) is represented in orange. Additionally, the cumulative impact of adding each feature is shown by the blue line. This visual representation helps to clarify which features significantly enhance the model's predictive accuracy and which do not.

This analysis concludes that optimal regression performance can be achieved using only four key features: the initial capacity at the start of the day (SOH₀), the mean value of the input profile (T_1), the operating strategy (α_{scen}), and the mean absolute value of the input profile (T_4). By eliminating the extraneous features, the likelihood of distortion from outlier estimations is reduced, and the simpler feature set enables faster training and prediction processes.

Adopting this streamlined approach, the design matrix now consists of just four columns, corresponding to the selected features. The regressors developed using this reduced set are referred to as *Low Dimensional Random Forests* (LDRF) and are represented as Model #4 in Fig. 5. Fig. 8 illustrates the comparison between the actual and predicted values of one-day capacity loss, showcasing that all data segments align

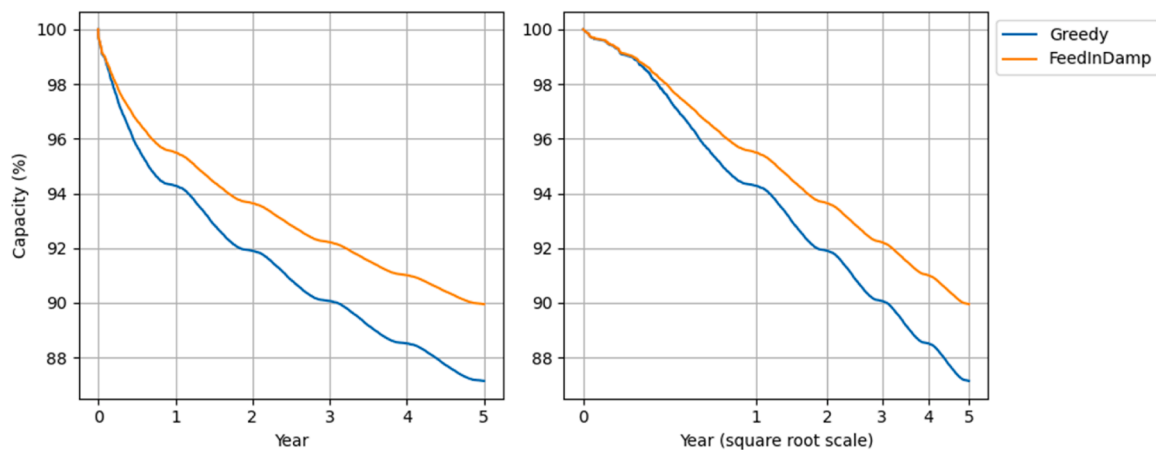


Fig. 6. Evolution of the BSS capacity for a randomly chosen household in the two possible operational strategies, plotted for a horizontal axis on a linear (left) and square root scale (right).

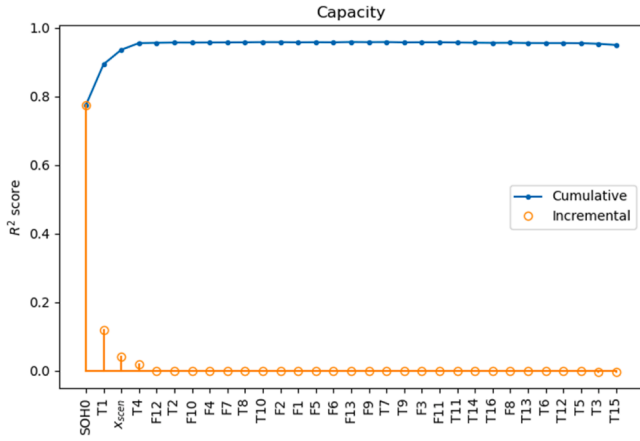


Fig. 7. Performance of a random forest regressor for an increasing number of features, according to the forward feature selection algorithm: incremental (orange) and cumulative contribution (blue).

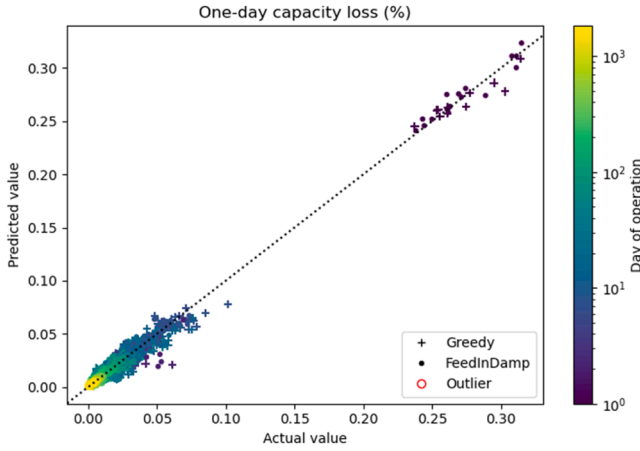


Fig. 8. Comparison between actual and predicted values of the capacity loss in one-day, for a short-term model based on a low-dimensional random forest (LDRF).

closely with the main diagonal, and notably, the previous outlier estimations are absent. The coefficient of determination for this model is an impressive $R^2 = 0.975$.

Long-term capacity estimation

Adapting Eq. (7) for the scenario of long-term capacity degradation, the estimated daily loss of BSS capacity, denoted as $\widehat{\Delta Q}_{loss}$, can be cumulatively summed to determine the total capacity loss by the end of the k -th day as follows:

$$\widehat{Q}_{loss}(k) = \sum_{u=1}^k \widehat{\Delta Q}_{loss}(u), \quad \forall k \leq k_{5y} = 5 \cdot 365, \quad (10)$$

An expression that remains valid for any day up to the last day of the five-year period, denoted as $k_{5y} = 5 \cdot 365$, allows us to estimate the total relative capacity at the end of the simulation period. The estimation can be performed by:

$$\widehat{Q}_{5y} = \frac{\widehat{Q}(k_{5y})}{Q(0)} = 1 - \widehat{Q}_{loss}(k_{5y}) = 1 - \sum_{u=1}^{k_{5y}} \widehat{\Delta Q}_{loss}(u), \quad (11)$$

The long-term estimation of battery capacity degradation, as detailed above, will be further explored in the subsequent subsections.

Five-years ahead capacity estimation

The comparison between the actual and predicted values of the BSS capacity at the end of a five-year period is illustrated in Fig. 9 (Model #5 in Fig. 5). Here, all data points—representing input profiles from the testing dataset—are closely aligned with the main diagonal, indicating a high level of accuracy in the long-term predictions. The density of dots in this figure is considerably lower compared to previous charts, such as Fig. 8. This difference arises because the points in Fig. 9 represent the cumulative effect of five years of capacity change, equating to 1825 (5×365) daily data points shown in Fig. 8.

The coefficient of determination for this regression, $R^2 = 0.932$, indicates a good but slightly lower performance compared to the metric obtained for daily capacity loss predictions. This difference is likely due to the accumulation of errors from daily estimates. In Fig. 8, data points positioned above the diagonal indicate an overestimation of SOH, while those below it represents an underestimation.

Improving estimations for feed-in-damping strategies

A detailed examination of Fig. 9 reveals that predictions for the feed-in damping strategies deviate more from the main diagonal compared to those from the greedy operational strategy. This initial visual observation is supported by a subsequent analysis of the statistical distribution of absolute estimation errors. These errors are defined as:

$$AE_i = |y_i - \widehat{y}_i|, \quad (12)$$

In this analysis, y_i and $\widehat{y}_i$ represent the actual and estimated values for the i -th instance, respectively. The distribution of absolute estimation errors, defined as $|y_i - \widehat{y}_i|$, for the two operational strategies is depicted in Fig. 10 using violin plots. These plots effectively illustrate the density and distribution of errors, highlighting the variance between the feed-in damping and greedy strategies.

As outlined in Section 2.1, the feed-in strategy relies on forecasting solar power generation. However, none of the features in either the high-dimensional or low-dimensional sets currently include information to estimate photovoltaic generation, which complicates accurate estimation of BSS capacity degradation under this strategy.

To address this limitation, adding the day of the year as a new fifth attribute to the low-dimensional dataset provides a straightforward solution. This attribute, ranging from 1 to 365, correlates with photovoltaic power generation, as demonstrated in Fig. 11. The small blue dots in the figure represent the average photovoltaic power generated on each day of the year. To smooth these data, a Savitzky-Golay filter [65] of third order with a window size of 30 days is applied, resulting in the smooth orange line shown in Fig. 11.

With the inclusion of the day of the year as the fifth feature (selected features + day), Model #6 in Fig. 5 can first estimate daily capacity

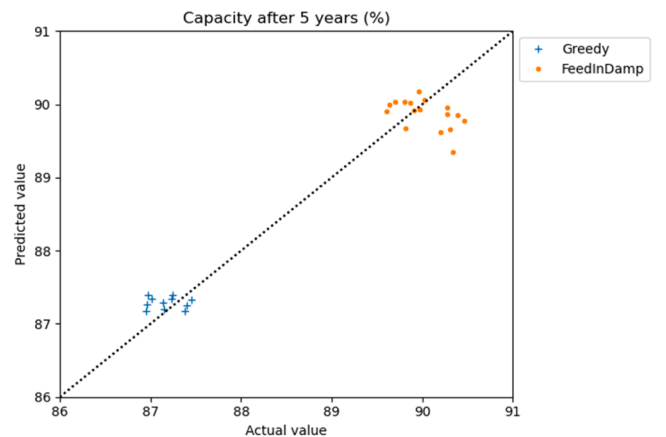


Fig. 9. Comparison between actual and predicted values of the BSS capacity after a five-years evolution.

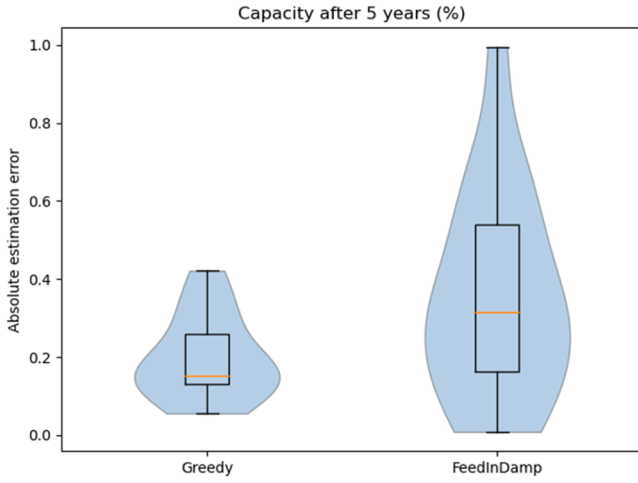


Fig. 10. Statistical distribution of the absolute estimation errors of the BSS capacity after a five-years evolution in two operational strategies. The blue shadow area represents the probability density function of the errors.

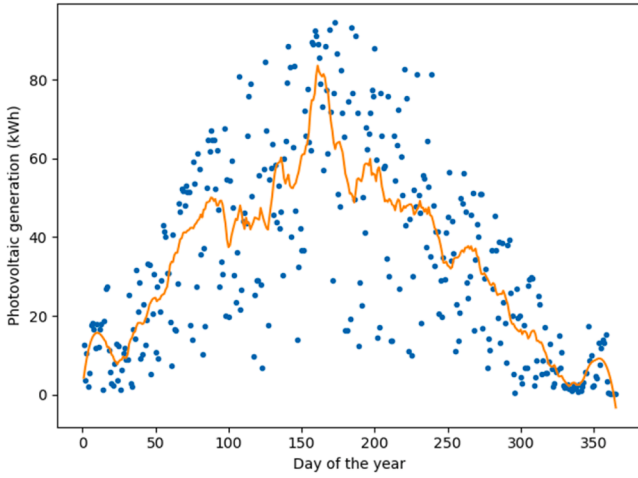


Fig. 11. Photovoltaic generation on each day of the year (small blue dots). A smoothing line, obtained using a Savitzky-Golay filter, is also plotted (orange line).

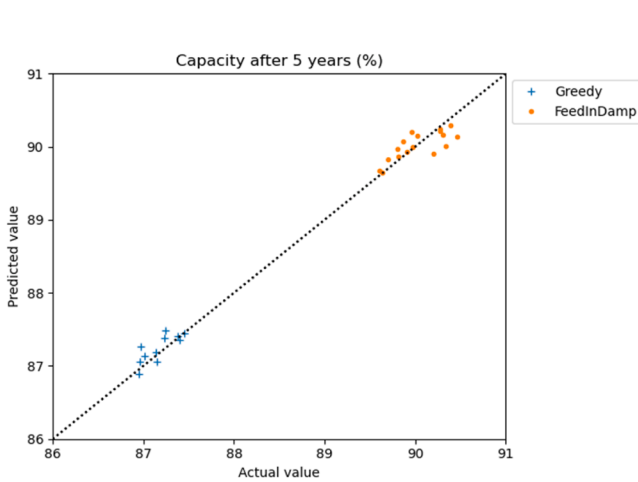


Fig. 12. Comparison between actual and predicted values of the BSS capacity after a five-years evolution, for a low-dimensional model which considers the day of the year.

decrease. These daily values can then be accumulated to determine the BSS capacity at the end of five years, as executed in Model #7 in Fig. 5. The comparison between actual and predicted long-term values is shown in Fig. 12, where all data points are now positioned closer to the main diagonal compared to those in Fig. 9. This adjustment results in similar regression performances across the two operational strategies. The coefficient of determination for this enhanced model is $R^2 = 0.988$, significantly higher than that achieved by the previous models.

The statistical distribution of the absolute estimation errors for this improved model is depicted in Fig. 13 using violin plots for each of the two operating strategies.

Machine learning-based simulator

The refined one-day ahead regressor (Model #6 in Fig. 5), utilizing a low-dimensional design matrix with five features (selected features + day), serves as the basis for an ML-based Battery Life Predictor. This tool, which is now publicly available [41], processes input power profiles (PV generation and residential demand) provided in a CSV file to estimate the SOH of a battery system over time. In Fig. 14, the ML-based simulation results for a randomly selected household are shown for two operational strategies. These results, represented by dashed lines, are compared against the SOH calculations from the SimSES framework, depicted with solid lines.

Continuous-time capacity estimation

In previous sections, we have estimated BSS capacity degradation up to five years, in a daily base, as this is the duration for which actual values are available. To extend this analysis, a continuous time long-term estimation of BSS capacity is derived in Appendix C. According to Eq. (18), it is possible to predict its Useful Life (UL), as illustrated in Model #9 in Fig. 5. The UL is defined as the time until the capacity falls below a certain predefined threshold, Q_{th} —typically 80% of the original capacity. Fig. 15 depicts this concept, showing the evolution of battery capacity for a mean household as a line, $\bar{Q}(t)$. The shadowed area around this line represents the estimation range, defined as $\bar{Q}(t) \pm 4\sigma_Q(t)$, where:

$$\bar{Q}(t) = \frac{1}{M} \sum_{i=1}^M \hat{Q}^{(i)}(t). \quad (13)$$

The UL, for $Q_{th} = 80\%$, is also indicated in Fig. 15. The RUL for an average household at a given time, can be expressed as $\overline{RUL}(t) = UL - t$, indicating a linear decrease over time. The profiles are categorized into

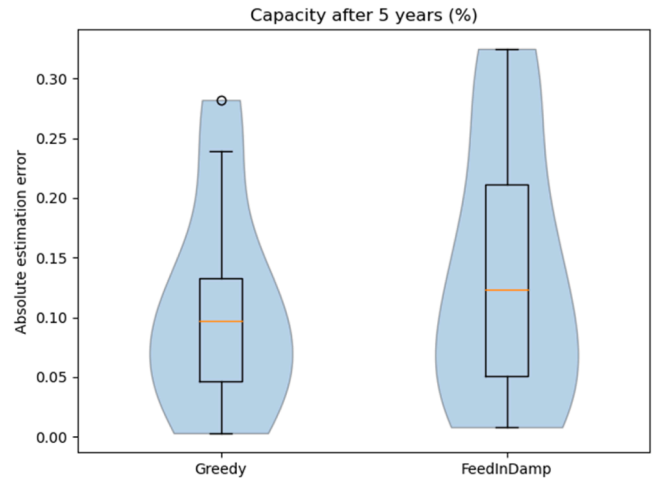


Fig. 13. Statistical distribution of the absolute estimation errors of the BSS capacity after a five-years evolution in two operational strategies, for a low-dimensional model which considers the day of the year.

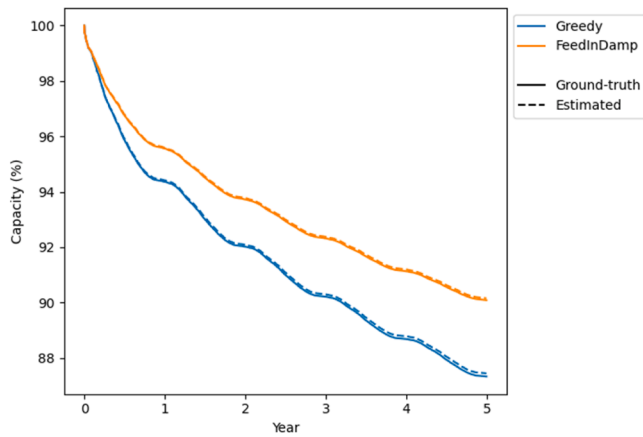


Fig. 14. ML-based simulation of an arbitrarily selected household demand. The simulation results (dashed lines) are compared with the SOH evolution calculated in SimSES (solid line).

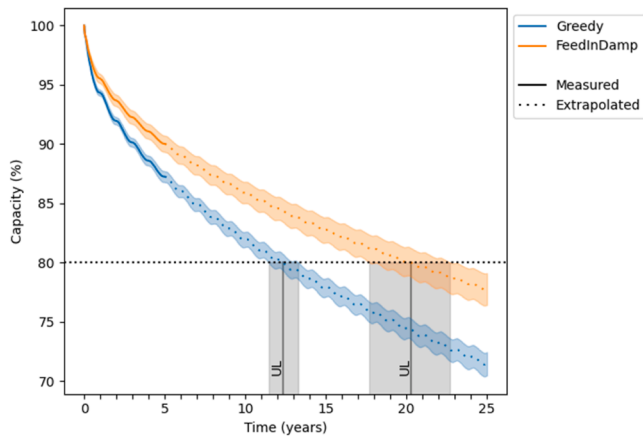


Fig. 15. Long-term continuous-time evolution of BSS capacity. The mean values up to five years (solid lines), its extrapolation beyond that period (dashed lines), and the 4σ -uncertainty (shadowed regions) are represented. Several concepts related to Useful Life are also indicated: capacity threshold (black dashed line), mean Useful Life (black solid line) and Useful Life's 4σ -uncertainty (gray shadowed area).

two groups based on their operational strategy. It can be seen that batteries operated under the "greedy" strategy have an expected useful life of 12.4 years, with a 4σ uncertainty of 0.9 years, resulting in a range of 11.5–13.3 years. Conversely, batteries utilizing a "feed-in damp" strategy exhibit a longer average useful life of 20.3 years, with an uncertainty of about 2.5 years, extending from 17.8 to 22.8 years.

Calendar and cycle aging

The analysis described in previous sections addresses battery degradation as a cumulative effect of various causes. For a more detailed examination, this degradation can be categorized into two primary causes: calendar aging, which results from the passage of time, and cycle aging, which results from battery usage. This refined analysis is facilitated by SimSES, which provides not only the total State of Health (SOH) but also its separate calendar and cycle components.

Using a similar methodological approach, a recursive forward feature selection algorithm was applied, identifying the same set of five most relevant features for this detailed analysis. These selected features are then used to characterize input profile segments, which train the corresponding disaggregated LDRF models. These models are capable of predicting daily calendar and cycle degradation.

The accumulated results of these daily degradations are used to estimate the disaggregated SOH evolution. Fig. 16 illustrates this evolution for an average household, showing how both types of aging contribute to overall battery degradation over time.

It can be observed from the analysis that approximately two-thirds of the SOH degradation is attributed to calendar aging, while one-third results from cycle aging.

Discussion

Summary of estimation performances

Referring back to Fig. 5, up to five sets of features (identified on the x-axis) have been proposed for characterizing input profiles. These feature sets were then utilized to train regressors responsible for predicting daily battery degradation (as shown in models 1, 2, 3, 4, and 6 in Fig. 5). The outcomes from these daily predictions serve as the basis for extending to long-term predictions (models 5, 7, 8, and 9 in Fig. 5), which are essentially accumulations of short-term predictive results. Consequently, for evaluating estimation performances, it suffices to focus solely on the short-term models, which aim for a one-day ahead prediction horizon.

The coefficient of determination, R^2 , is employed to evaluate the performance of models using five distinct sets of features: *Residual*; *Residual+SOH*; *Residual+SOH+Strategy*; *Selected Features*; and *Selected Features+Day*. The results, displayed in Fig. 17 (left panel), demonstrate that the accuracy of capacity estimation significantly improves as the complexity of the feature set increases.

On the other hand, the performances of the disaggregated SOH predictions using the set of *Selected Features + Day* have been evaluated. The outcomes of this analysis are presented in Fig. 17 (right panel), indicating that estimating cycle aging is significantly more challenging than estimating the corresponding calendar aging.

Computing effort of machine learning estimations

While prediction performance is the primary criterion for selecting a machine learning algorithm, it is also important to consider the computational costs of these estimations. These costs can be categorized into three distinct areas: the time required to extract the set of features from a five-year input profile; the time to train the model using 80 % of the available dataset, which includes 216,080 one-day length segments; and the time needed to predict the BSS capacity after processing a five-year input profile.

The processing times are summarized in Table 2 and were obtained using a desktop computer equipped with an Intel® Core™ i7-11700

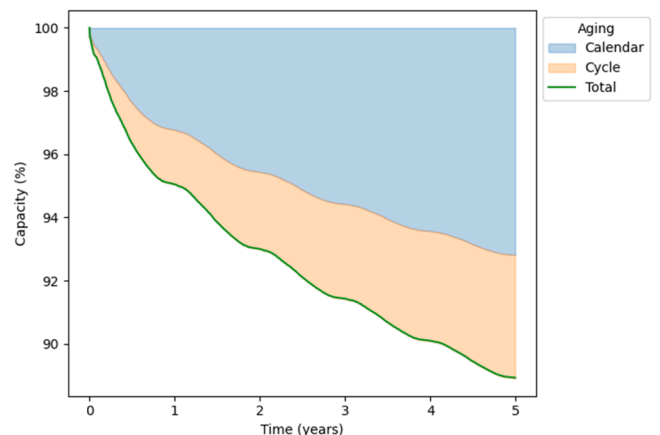


Fig. 16. Capacity evolution disaggregated by its two causes: calendar and cycle aging. SOH values are estimated using the ML approach.

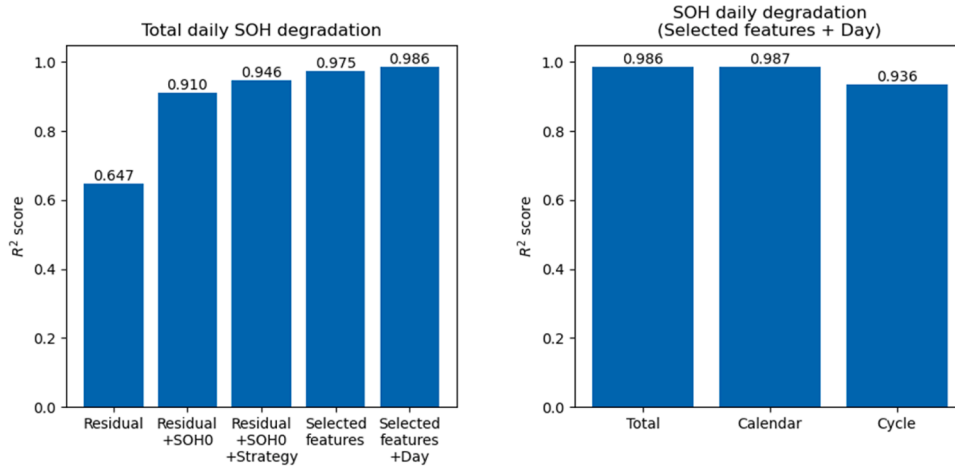


Fig. 17. Comparison of performances for estimating SOH daily degradation: (left panel) total degradation for different set of features; (right panel) disaggregated degradation for a *Selected features + Day* set of features.

Table 2

Computing effort required to extract the features characterizing a five-year input profile.

Model	Set of features	Extraction (ms)	Training (s)	Prediction (ms)
HDRF	Residual	610	240	6.8
	Residual + SOH_0			
	Residual + SOH_0 + Strategy			
LDRF	Selected features	31	28	6.5
	Selected features + Day			

processor at 2.50 GHz, 64 GB of RAM, and a 2 TB Solid-State Disk (SSD). The tasks of feature extraction, model training, and capacity prediction were conducted using Python 3.9 on a Windows 11 operating system. For machine learning tasks, the scikit-learn library, version 1.0, was utilized.

Extraction times vary depending on the number and type of features to be extracted from the input profile. For the first three feature sets, a total of 29 features are extracted, as defined in Table A-1 and Table A-2. Conversely, only 2 features (T_1 and T_4) are extracted for the set labeled "Selected features." It is important to note that features such as SOH_0 , Strategy, and Day do not require extraction from the input profile as they are derived directly from known parameters or settings.

Training and prediction times are influenced not only by the number of features but also by the complexity of the model, which includes the number of trees in the random forest and the maximum depth of these trees—essentially, the hyperparameters of the model. Additionally, the specific datasets used for training and prediction also play a significant role. Generally, training times are considerably longer than prediction times, often ranging from 10^4 to 10^5 times longer. However, it is important to note that the training process is conducted only once, when the model is initially built.

Finally, the approximate 7 ms required for an ML estimation of the SOH should be considered in comparison to the 54.9 minutes required to process the equivalent five-year input profile using the SimSES simulation framework [44]. This comparison highlights the substantial time efficiency of ML models. The SimSES simulation was conducted on a desktop computer equipped with an Intel® Xeon® W-2295 CPU at 3 GHz, 256 GB of RAM, and a 315 GB Solid-State Disk (SSD).

Physical interpretation of low-dimensional features

The dimensionality reduction process used to estimate daily degradation of BSS capacity identified several key features as most relevant:

the capacity at the beginning of the day (T_1), the operating strategy, and T_4 . Let's explore the underlying reasons why these specific features emerged as critical in this context.

According to Eq. (4), capacity degradation in battery storage systems can be modeled by the non-linear function, $Q_{loss}(t) \approx K_Q \sqrt{t}$. Thus, its gradient (the incremental capacity loss on the k -th day) is dependent on the operating point (the SOH at the beginning of that day), denoted as $SOH(k)$. This dependency highlights why $SOH(k)$ is the most relevant feature for predicting daily capacity degradation.

To illustrate the dependency of daily capacity loss ΔQ_{loss} on various variables, we have visualized the dataset in a five-dimensional space. Each dot in this space represents an instance of ΔQ_{loss} as a function of four variables: initial SOH (SOH_0), mean residual power (T_1), operating strategy (x_{scen}), and mean absolute residual power (T_4), formulated as $f_Q(SOH_0, T_1, x_{scen}, T_4)$. To simplify visualization, we slice this five-dimensional cloud of dots with a hyperplane at the mean values of T_1 and T_4 , yielding a three-dimensional (3D) representation expressed as $\Delta Q_{loss} = f_Q(SOH_0, \bar{T}_1, x_{scen}, \bar{T}_4)$.

Further, this 3D cloud is projected onto the $(\Delta Q_{loss}, SOH_0)$ plane, resulting in a two-dimensional (2D) representation, $\Delta Q_{loss} = f_{Qpr}(SOH_0, \bar{T}_1, \bar{T}_4)$. The data points in this 2D representation are displayed in the left panel of Fig. 18. Additionally, these data points are optimally fitted to the aging model expressed in Eq. (4), and depicted as an orange line.

According to this aging model, the incremental capacity loss on the k -th day is influenced by parameters that depend on the values of ambient temperature, SOC, C-rate, DOD, and the number of FEC on that particular day. Notably, except for ambient temperature, the other four parameters are influenced by the input profile. For example, the mean values of SOC or DOD for a day are higher when the mean residual power, denoted as T_1 , increases. Conversely, C-rate and FEC are influenced by the residual power irrespective of its sign, relying instead on its mean absolute value, T_4 .

The impact of these relationships is visualized in Fig. 18. The data points for $\Delta Q_{loss} = f_{Qpr}(SOH_0, \bar{T}_1, \bar{T}_4)$ are displayed in the central panel of Fig. 18, along with a linear fit represented by an orange line. Additionally, the function $\Delta Q_{loss} = f_{Qpr}(SOH_0, \bar{T}_1, \bar{T}_4)$ and its linear fitting are presented in the right panel of Fig. 18. In this panel, the dependence on T_4 is less pronounced, indicating its reduced contribution to the estimation of capacity degradation, as evidenced by the findings depicted in Fig. 7.

The evolution of BSS capacity exhibits a seasonal behavior, as observed in Fig. 6 and Fig. 15, with a higher degradation rate in summer compared to winter. This seasonal variation is due to the increased daily

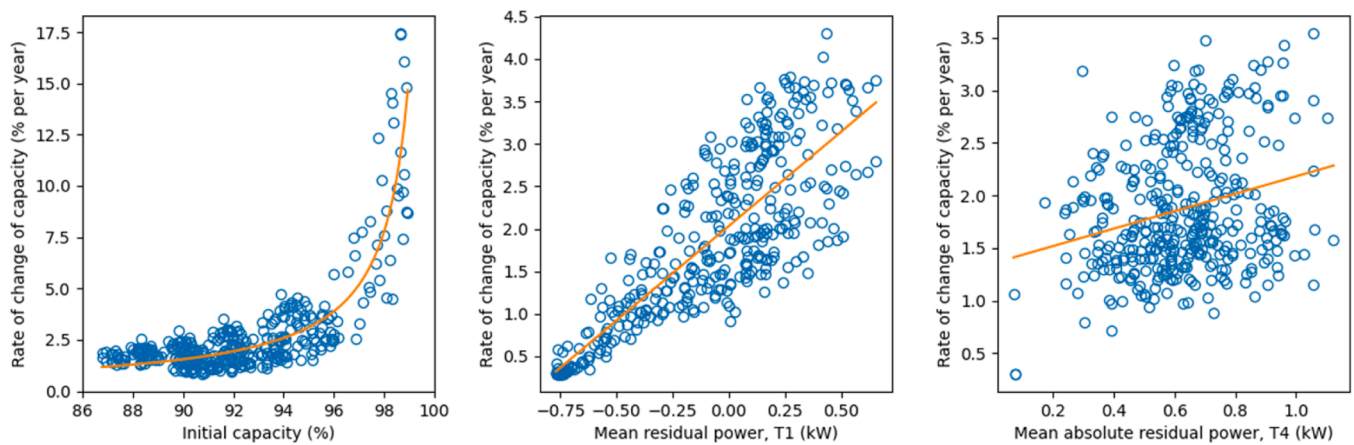


Fig. 18. Relationship between daily loss of capacity and initial capacity at the beginning of the day (left), mean residual power T_1 (center), and mean absolute residual power T_4 (right).

photovoltaic generation during summer, resulting in a positive mean residual power, whereas in winter, the mean residual power is negative. Consequently, the battery's SOC is higher in summer, while the DOD is lower. Additionally, the mean absolute residual power is higher in summer, leading to increased C-rate and FEC.

Fig. 19 illustrates these seasonal effects, with the daily change in capacity plotted over time in the top panel. The middle and bottom panels show the values of T_1 and T_4 , respectively. All three panels clearly demonstrate the seasonal behavior, highlighting the influence of varying photovoltaic generation and residual power on the degradation rate throughout the year.

Conclusions

This research demonstrates the effectiveness of machine learning techniques in predicting the SOH of BSS used in SCI applications. By leveraging photovoltaic generation and household power demand

profiles, our data-driven approach replicates state of health (SOH) estimations from circuit-equivalent simulations with remarkable accuracy, achieving an average error of only 0.1 % for five-year predictions. Furthermore, it delivers results approximately six orders of magnitude faster than traditional simulations, completing predictions in milliseconds compared to 54.9 minutes.

The machine learning models, implemented through random forest regressors, predict daily SOH changes and aggregate these to forecast long-term effects. These models use a low-dimensional input profile comprising the mean and mean absolute values of generated and demanded power, along with the initial daily SOH, operating strategy, and day of the year. This approach allows for the determination of the useful life (UL) of BSS, with an 80 % capacity threshold, predicting approximately 13 years under a "greedy" operational strategy and around 20 years with a "feed-in damp" strategy.

Our detailed analysis reveals that calendar aging is more predictable than cycle aging. The open-source MLBatLife SOH prediction tool developed in this study offers significant utility for battery manufacturers and installation engineers, enabling them to estimate the need for system oversizing in specific usage scenarios to meet expected lifetimes before installation. This tool also facilitates economic analyses by providing accurate UL estimates, which are crucial for amortization calculations.

Future research should focus on expanding this methodology to other battery use cases, such as peak shaving or frequency containment reserves, and adapting it to various battery chemistries and storage dimensions. Moreover, it is essential to validate these findings against real-world data beyond the SimSES simulations to ensure robustness and practical applicability.

CRedit authorship contribution statement

Schroeder Benedikt: Writing – review & editing, Software. **Luque Joaquin:** Writing – original draft, Software, Methodology. **Hesse Holger:** Writing – review & editing, Validation, Supervision, Conceptualization. **Leon Carlos:** Writing – review & editing, Validation, Resources. **Heidarabadi Housman:** Writing – review & editing, Writing – original draft, Investigation. **Carrasco Alejandro:** Writing – review & editing, Writing – original draft, Validation.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Carlos Leon reports financial support was provided by Spain Ministry of Science and Innovation. Holger Hesse reports financial support was

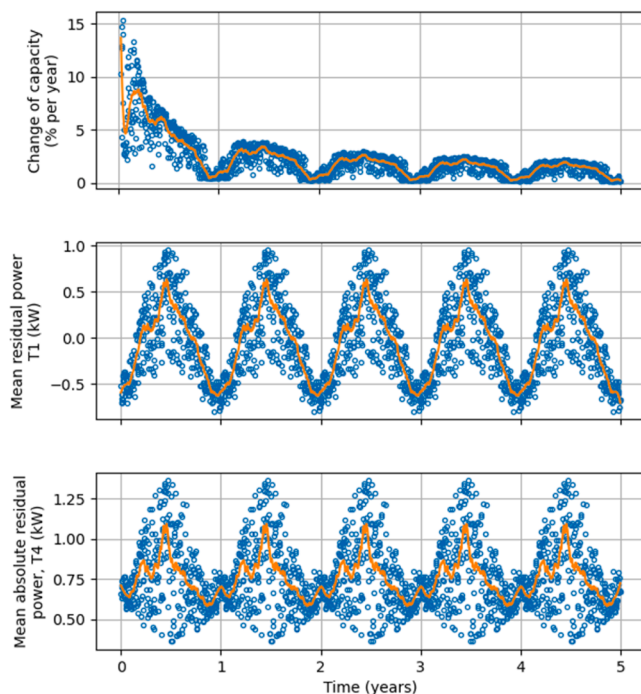


Fig. 19. Seasonal behavior of BSS degradation: daily change of capacity (top), mean residual power (middle), and mean absolute residual power (bottom).

provided by German Federal Ministry for Economic Affairs and Energy. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Avail ability

The power demand of 74 families has been recorded by the HTW Berlin for one year (2010) [42]: <https://solar.htw-berlin.de/elektrische-lastprofile-fuer-wohngebaeude>

The SimSES software is available as open source as indicated in Appendix E of [44].

Appendix A. . Features defining an input profile

A one-day input profile segment, containing $N = 86400$ values sampled at one-second intervals, can be summarized using a set of 29 generic features derived through statistical analysis in both the time and frequency domains. From the time-domain samples, up to 16 features are computed, as described in Table A-1, where the term s_i corresponds to the i -th sample of the segment.

Table A-1

Features extracted from the time-domain values of a segment

Name	Meaning	Expression	Name	Meaning	Expression
T_1	Mean	$\frac{1}{N} \sum_{i=1}^N s_i$	T_9	Minimum absolute value	$\min_i s_i $
T_2	Standard deviation	$\sqrt{\frac{1}{N-1} \sum_{i=1}^N (s_i - T_1)^2}$	T_{10}	Peak-to-peak value	$T_8 - T_9$
T_3	Square root amplitude	$\left(\frac{1}{N} \sum_{i=1}^N \sqrt{ s_i } \right)^2$	T_{11}	Waveform index	$\frac{T_2}{T_4}$
T_4	Absolute mean value	$\frac{1}{N} \sum_{i=1}^N s_i $	T_{12}	Peak index	$\frac{T_8}{T_2}$
T_5	Skewness	$\frac{1}{N} \sum_{i=1}^N s_i^3$	T_{13}	Pulse index	$\frac{T_8}{T_4}$
T_6	Kurtosis	$\frac{1}{N} \sum_{i=1}^N s_i^4$	T_{14}	Margin index	$\frac{T_8}{T_3}$
T_7	Variance	$\frac{1}{N} \sum_{i=1}^N s_i^2$	T_{15}	Skewness index	$\frac{T_5}{(\sqrt{T_7})^3}$
T_8	Maximum absolute value	$\max_i s_i $	T_{16}	Kurtosis index	$\frac{T_6}{T_7^2}$

By applying the Fast Fourier Transform (FFT) to the N samples of a one-day segment, we obtain a representation in the frequency domain that also consists of N harmonic components. The frequency of the last component is defined as $f_N = f_s$, where f_s is the sampling rate. For this dataset, $f_s = 1\text{Hz}$. Therefore, the frequency of the i -th harmonic is calculated as $f_i = i \cdot f_s / N = i / N \text{ Hz}$, and the amplitude of this harmonic is denoted by S_i . According to the sampling theorem, only the first $M = N/2$ harmonic components are considered significant. The 13 frequency-domain features derived from these components are listed in Table A-2.

Table A-2

Features extracted from the frequency-domain values of a segment

Name	Meaning	Expression	Name	Meaning	Expression
F_1	Spectral mean	$\frac{1}{M} \sum_{i=1}^M S_i$	F_8	Quartic spectral centroid	$\sqrt{\frac{\sum_{i=1}^M f_i^4 S_i}{\sum_{i=1}^M f_i^2 S_i}}$
F_2	Spectral variance	$\frac{1}{M-1} \sum_{i=1}^M (S_i - F_1)^2$	F_9		$\frac{\sum_{i=1}^M f_i^2 S_i}{\sqrt{\left(\sum_{i=1}^M f_i^4 S_i \right) \left(\sum_{i=1}^M S_i \right)}}$
F_3	Spectral skewness	$\frac{\sum_{i=1}^M (S_i - F_1)^3}{M(\sqrt{F_2})^3}$	F_{10}	Spectral dispersion index	$\frac{F_6}{F_5}$
F_4	Spectral kurtosis	$\frac{\sum_{i=1}^M (S_i - F_1)^4}{MF_2^2}$	F_{11}		$\frac{\sum_{i=1}^M (f_i - F_5)^3 S_i}{MF_6^3}$
F_5	Spectral centroid	$\frac{\sum_{i=1}^M f_i S_i}{\sum_{i=1}^M S_i}$	F_{12}		$\frac{\sum_{i=1}^M (f_i - F_5)^4 S_i}{MF_6^4}$
F_6	Spectral dispersion	$\sqrt{\frac{\sum_{i=1}^M (f_i - F_5)^2 S_i}{M}}$	F_{13}		$\frac{\sum_{i=1}^M \sqrt{ f_i - F_5 } S_i}{M\sqrt{F_6}}$
F_7	Quadratic spectral centroid	$\sqrt{\frac{\sum_{i=1}^M f_i^2 S_i}{\sum_{i=1}^M S_i}}$			

Preliminary high-dimensional models

Basic set of features

The basic set of features, referred to as "*residual*," consists of 29 attributes detailed in the time domain (Table A-1) and frequency domain (Table A-2). Utilizing these features to characterize input profiles, a random forest regressor (model #1 in Fig. 5) was trained. The effectiveness of this model is illustrated in Fig. B-1, where each dot represents a one-day segment. The capacity loss for each segment is depicted as a percentage of the original, non-degraded capacity.

The coefficient of determination for this regression, $R^2 = 0.663$, indicates a relatively low performance, which corresponds with the dispersion of the dots away from the main diagonal in Fig. B-1. The colors of the dots, represented on a logarithmic scale, denote the number of days since the beginning of the battery's use. Notably, the dots situated farthest to the right, which depict the highest daily capacity losses, align with the first day of operation. A detailed physical explanation of this phenomenon is provided in Section 4.3. Additionally, some outliers are marked with red circles in the graph, with their causes discussed in Appendix D.

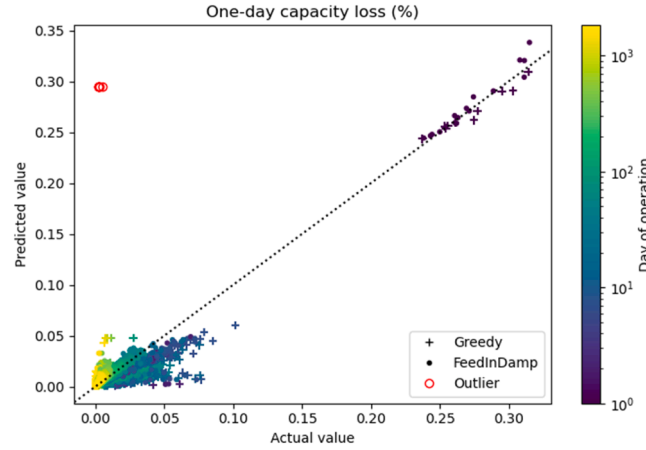


Fig. B-1. Comparison between actual and predicted values of the capacity loss in one-day, for a basic short-term model. Colors correspond to the day of operation

Non-linear time evolution

It can be observed that the previous model tends to underestimate the one-day capacity loss. This underestimation stems from the model's reliance solely on the residual power, coupled with the assumption that the daily capacity loss remains constant regardless of the absolute value of the BSS capacity. This implies a linear degradation of BSS capacity over time, assuming a constant degradation rate for any given input profile. However, this assumption has been disproven by the data presented in Fig. 6, which illustrates that the degradation rate is not constant.

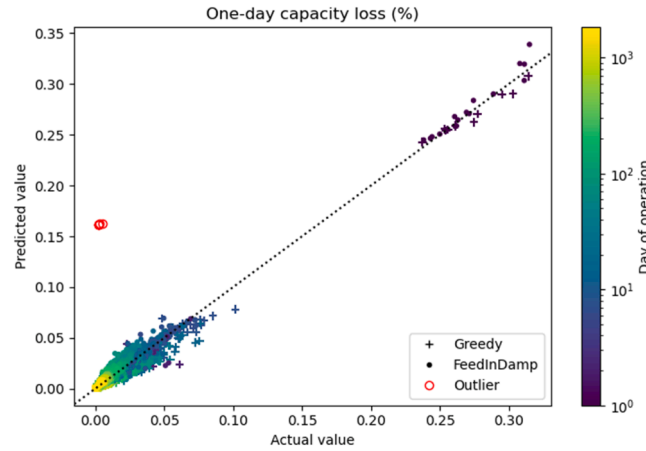


Fig. B-2. Comparison between actual and predicted values of the capacity loss in one-day, for a short-term model which considers the BSS capacity at the beginning of the day

To account for the non-linearity in capacity degradation, the capacity at the beginning of the day (SOH_0) has been added as a new 30th column to the design matrix, forming the augmented feature set (*residual*+ SOH_0). This modification is implemented in model #2, depicted in Fig. 5. The results, illustrated in Fig. B-2, show that most of the data points (segments) align closely with the main diagonal, indicating a significant improvement in prediction accuracy. This revised model achieves an enhanced coefficient of determination, $R^2 = 0.910$, reflecting substantial improvements in model performance.

Considering operation strategy

While the first two models considered only the residual power profile, they did not account for whether the Battery Storage System (BSS) was

operating under "greedy" or "feed-in damp" strategies, as described in Section 2.1. By integrating this operational strategy information into the design matrix (*residual*+*SOH*₀+*strategy*) as a new 31st column, an improvement in performance is anticipated. The results for this enhanced regressor (model #3 in Fig. 5) are presented in Fig. B-3, where each dot represents a one-day segment and the capacity loss is expressed as a percentage of the original, non-degraded capacity. Compared to the preliminary models (details in Appendix B), the data points for this regressor are noticeably closer to the main diagonal, indicating a significantly higher coefficient of determination, $R^2 = 0.946$. However, the presence of five outliers in the graphic is noteworthy, and these have been zoomed in for clearer examination.

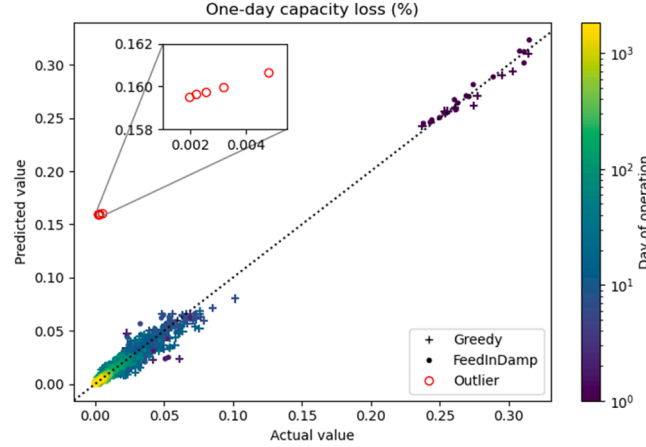


Fig. B-3. Comparison between actual and predicted values of the capacity loss in one-day, for a short-term model which considers the BSS operating strategy. The estimation outliers have been zoomed in

Deriving continuous-time models

In Section 3.3, we have estimated BSS capacity degradation up to five years, as this is the duration for which actual values are available. To project these predictions further, such as k_e days beyond the fifth year, the LDRF model can be applied. This model utilizes the set of five features from the i -th input profile (household) at the extended date, denoted as $x^{(i)}(k_{5y} + k_e)$, allowing for:

$$\widehat{\Delta Q}_{\text{loss}}^{(i)}(k_{5y} + k_e) = \text{LDRF}[x^{(i)}(k_{5y} + k_e)]. \quad (14)$$

The expression $x^{(i)}(k_{5y} + k_e)$ denotes a five-valued vector representing the i -th input profile (residual power of the i -th household), k_e days after the end of the fifth year. This set of five features employed in the LDRF model for extended predictions can be expressed as follows:

$$x^{(i)}(k_{5y} + k_e) = [Q^{(i)}(k_{5y} + k_e - 1), T_1^{(i)}(k_d), x_{\text{scen}}^{(i)}, T_4^{(i)}(k_d), k_d]. \quad (15)$$

The set of five features for the extended prediction period utilizes the modulus operation to account for time cyclicity, where $k_d = (k_{5y} + k_e) \bmod k_{1y} = k_e \bmod k_{1y}$. This operation ($a \bmod b$) returns the remainder after dividing a by b , allowing the model to treat extended days k_e as cyclic within a typical one-year period. The values of capacity degradation can be iteratively computed starting with:

$$\widehat{\Delta Q}_{\text{loss}}^{(i)}(k_{5y} + 1) = \text{LDRF}[Q^{(i)}(k_{5y}), T_1^{(i)}(1), x_{\text{scen}}^{(i)}, T_4^{(i)}(1), 1]. \quad (16)$$

Then, following Eq. (10), the capacity at the end of the k -th day, which may extend beyond the initial five-year period, can be calculated using the expression:

$$\widehat{Q}^{(i)}(k) = Q(0) \left(1 - \sum_{u=1}^k \widehat{\Delta Q}_{\text{loss}}^{(i)}(u) \right), \quad \forall k. \quad (17)$$

Additionally, by applying Eq. (8) specifically to BSS capacity estimation (as depicted in Model #8 in Fig. 5), the value for the i -th output profile at any time t can be estimated. This is achieved by interpolating between the capacity values at the beginning and the end of the corresponding day. The interpolation is performed as follows:

$$\widehat{Q}^{(i)}(t) = \widehat{Q}^{(i)}(k-1) + \frac{\widehat{Q}^{(i)}(k) - \widehat{Q}^{(i)}(k-1)}{t_k - t_{k-1}}(t - t_{k-1}), \quad (18)$$

where $t_k - 1 \leq t \leq t_k$.

On the other hand, the error in the estimation for the i -th output profile at any time t , denoted as $\varepsilon_Q^{(i)}(t)$, is defined by the equation $\varepsilon_Q^{(i)}(t) = \widehat{Q}^{(i)}(t) - Q^{(i)}(t)$. The mean value of this estimation error is then calculated as follows:

$$\bar{\varepsilon}_Q(t) = \frac{1}{M} \sum_{i=1}^M \varepsilon_Q^{(i)}(t), \quad (19)$$

Following the calculation of the mean, the standard deviation of the estimation error, $\sigma_{\varepsilon_Q}(t)$, is computed as follows:

$$\sigma_{\varepsilon_Q}(t) = \sqrt{\frac{1}{M-1} \sum_{i=1}^M \left[\varepsilon_Q^{(i)}(t) - \bar{\varepsilon}_Q(t) \right]^2}, \quad (20)$$

where M is the total number of output profiles.

The estimated values of $\hat{Q}^{(i)}(t)$ can be computed for any time t , even beyond the fifth year, using Eqs. (17) and (18). However, the actual values, $Q^{(i)}(t)$, and consequently, the estimation errors $\varepsilon_Q^{(i)}(t)$, are available only for the duration of the five-year period. To address estimation errors beyond this period, we must extrapolate from the known values. Fig. C-1 illustrates both the measured and extrapolated standard deviations of these estimation errors.

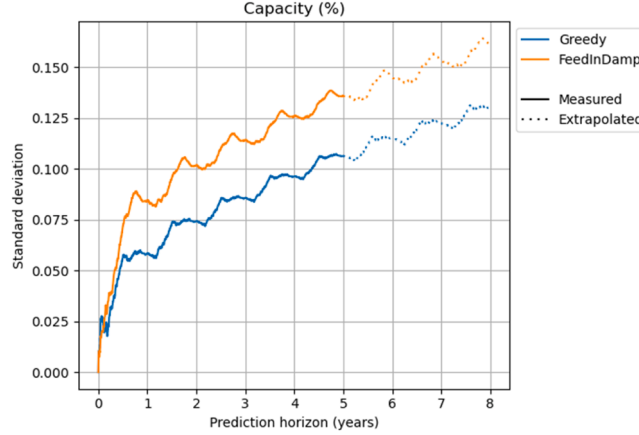


Fig. C-1. Standard deviation of the estimation error. Values beyond the fifth year are extrapolated

Explaining outliers in high-dimensional models

A detailed analysis of the outliers identified in the high-dimensional short-term capacity estimations (model #3 in Fig. 5) was conducted by examining the 31 features corresponding to each segment in the testing dataset. For more accurate comparisons, the feature values were preprocessed using z-score normalization [66]. The distribution of these normalized features is depicted in the boxplots shown in Fig. D-1. Notably, in this graphic, the values of the normalized features for the outlier estimations are marked as small circles. A key observation from this analysis is that the five outliers exhibit identical values across all features except one: the capacity at the beginning of the day. This indicates that these outliers originate from the same daily residual power profiles but occur in different years, when the BSS capacity varies (the input profiles are one-year recordings repeated five times).

This observation indicates that all the outliers originate from identical daily residual power profiles, but at different years. The only difference in these cases is the BSS capacity, which varies due to the aging process. Each input profile corresponds to one-year recordings that are repeated over a five-year period.

For an improved diagnosis of the causes of prediction errors, outlier estimations associated with extreme feature values are highlighted in red. This visualization occurs when the feature values are at the edges of their respective statistical distributions, which is observed for features such as T_6 (kurtosis), T_9 (minimum absolute value), and F_2 (spectral variance). The operational strategy (x_{scen}), having only one of two possible values, consistently appears at the edge of its distribution and is thus excluded from this detailed analysis.

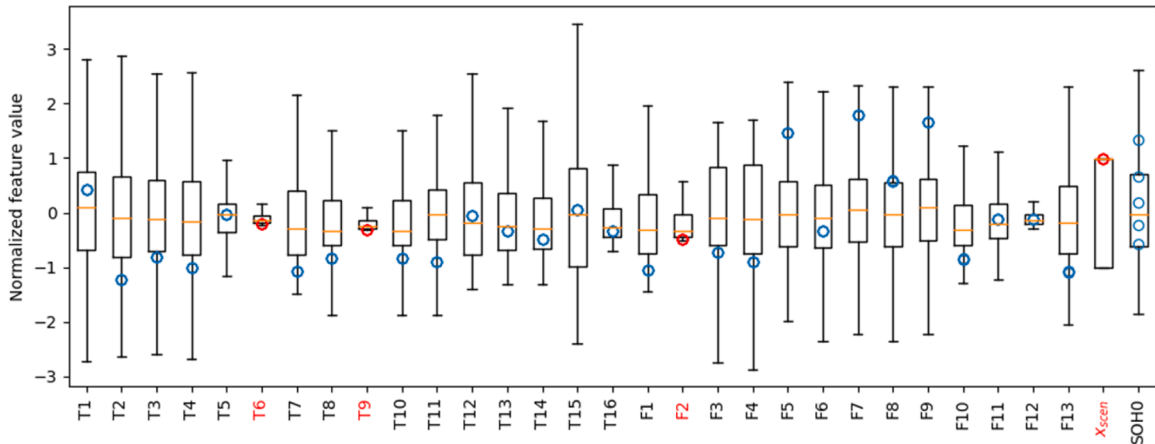


Fig. D-1. Boxplots showing the distribution of each z-score normalized feature in the testing dataset. The values for the outlier estimations are marked with a small circle (in red color if they are in the edge of the boxplot)

Following the methods outlined in Section 2.4, the importance of each feature in influencing the outcome of the random forest regressor has been meticulously analyzed. The importance rankings are displayed in descending order in Fig. D-2. Features associated with extreme values in the esti-

mation outliers are highlighted in red for clarity. Given that the operational strategy always registers as an extreme value, it is typically disregarded in this context. Consequently, the most significant feature contributing to the outliers, marked in red, is T_9 (minimum absolute value), indicating that this feature is primarily responsible for the observed anomalies.

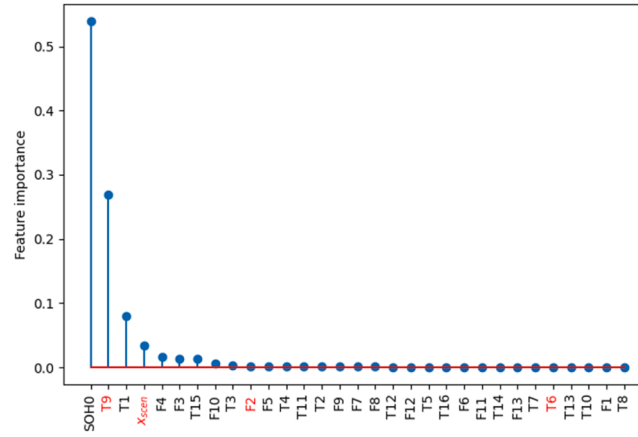


Fig. D-2. Feature importance as they are obtained for the random forest algorithm. Features with extreme values for the estimation outliers are written in red color

The features listed in Appendix A, which are used to characterize an input profile, consist of generic statistical summaries from both the time and frequency domains. Their generic nature indicates that they were not specifically tailored for this particular application—estimating the State of Health (SOH) in Battery Storage Systems (BSS). As a result, while some features may prove highly significant, others could be relatively inconsequential or even meaningless.

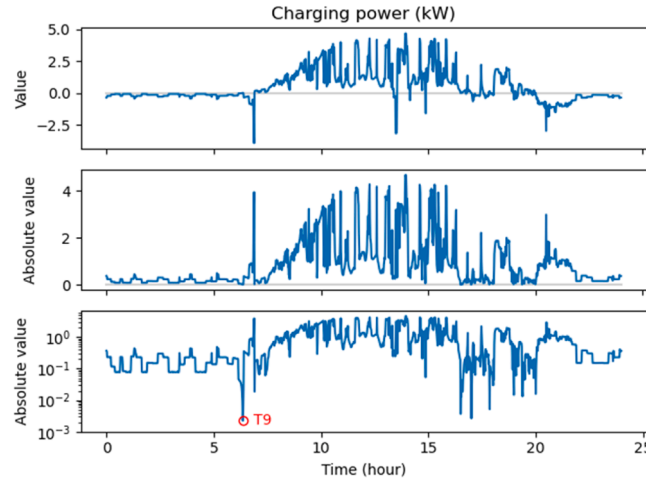


Fig. D-3. Example of the residual power input profile for a randomly chosen day: original value (top); absolute value in logarithmic scale (bottom). The value of T_9 is marked with a small red circle

A closer examination of the T_9 feature, which represents the minimum absolute value of the residual power, reveals that it holds little physical relevance in the context of this application. Consider the residual power for a randomly selected day and household, illustrated in the upper part of Fig. D-3. The minimum absolute value of this profile likely occurs at moments very close to zero, typically near where the power curve crosses the horizontal axis. The middle part of Fig. D-3 shows the absolute values of the residual power, with many points closely approaching the horizontal axis, making it challenging to pinpoint the exact minimum. For a more distinct visualization, the absolute residual power is plotted on a logarithmic scale in the lower part of Fig. D-3, where the minimum value is highlighted with a small red circle. This point corresponds to the value of T_9 for that specific segment.

From the analysis provided, it is evident that the T_9 feature, representing the minimum absolute value of the residual power, lacks relevant meaning in this application. Consequently, it must be concluded that the random forest algorithm erroneously identified T_9 as a significant feature, which has contributed to the estimation outliers. One potential solution to mitigate this issue could involve training the model with a larger dataset. A more extensive dataset might prevent the algorithm from selecting T_9 as a significant feature. However, given the constraints of the current dataset, alternative solutions need to be explored to address this issue.

After this analysis, it remains unclear why the training process identified the "minimum value" (T_9) as a highly important feature and why these values contributed to the outliers. However, the objective of this appendix is not to determine why T_9 holds such high importance but to emphasize that features lacking meaningful physical interpretation, such as T_9 , should be removed. This observation underscores the importance of applying the sequential forward feature selection algorithm described in Sections 2.4 and 3.2.

Acronyms and mathematical notation

To enhance the readability of the document, a list of all acronyms is provided in Table E-1. Additionally, the mathematical notation is outlined in

Table E-2. Finally, the meanings of the mathematical symbols are provided in Table E-3.

Table E-1
Acronyms used throughout the article

Acronym	Meaning	Acronym	Meaning
BMS	Battery Management System	ML	Machine Learning
BSS	Battery Storage Systems	PV	Photovoltaic
CPU	Central Processing Unit	RAM	Random Access Memory
DNN	Deep Neural Network	RF	Random Forest
DOD	Depth of Discharge	RNN	Recurrent Neural Network
EV	Electric Vehicle	RUL	Remaining Useful Life
FEC	Full Equivalent Cycles (charging-discharging)	SCI	Self Consumption Increase
FFT	Fast Fourier Transform	SOC	State of Charge
HDRF	High-Dimensional Random Forest	SOH	State of Health
LDRF	Low-Dimensional Random Forest	SSD	Solid-State Disk
LSTM	Photovoltaic	UL	Useful Life

Table E-2
Mathematical superscripts, subscripts, and overhead notations

Term	Meaning	Term	Meaning
$\cdot(i)$	i -th instance in a dataset	$\cdot 5y$	Value after a five-year period
$\cdot(t)$	Value of at time t	$\cdot e$	Value at the e -th day (after 5 years)
$\cdot(k)$	Value at the end of the k -th day	$\cdot d$	Value at the d -th day of the current year
$\cdot_k$	Value at the end of the k -th day	$\hat{\cdot}$	Estimated value
$\cdot 1y$	Value after a one-year period	$\bar{\cdot}$	Mean value

Table E-3
Mathematical symbols

Term	Meaning	Term	Meaning
AE	Absolute error	Q_{loss}^{cyc}	Capacity loss due to cycle aging
d	Number of components in x	Q_{th}	Coefficient of determination
d_z	Number of components in z	R^2	Coefficient of determination
F_j	j -th frequency-domain feature	t	Time
k	k -th day	T_j	j -th time-domain feature
K_{SOC}^{cal}	Calendar-SOC aging constant	x	Feature vector
K_T^{cal}	Calendar-temperature aging constant	X	Design matrix
K_{Crate}^{cyc}	Cycle-Crate aging constant	x_{scen}	Value of the “operating strategy” feature
K_{DOD}^{cyc}	Cycle-DOD aging constant	y	Target variable
K_T^{cyc}	Cycle-temperature aging constant	z	Low-dimensional feature vector
Q	Capacity of the BSS	ΔSOH	Incremental SOH
Q_{loss}	Capacity loss	ε_Q	Battery capacity estimation error
Q_{loss}^{cal}	Capacity loss due to calendar aging	σ_{ε_Q}	Standard deviation of ε_Q

Data availability

We have shared the link to my data in the manuscript

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